PROJECT AND TEAM INFORMATION

Project Title

Smart Campus Navigation System

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

Have you ever visited a college for the first time and had no idea where to go? I remember facing this exact situation on my own first day of college that I got so lost trying to find my classroom that I almost walked into the wrong building entirely. Big campuses usually have so many buildings, classrooms, labs, hostels, canteens, offices, and it is easy to get lost, especially on your first day. You end up asking random people for directions, and even then you might take a longer route without realizing it.

That is the problem which we wanted to solve with our project. What if there was a system that could tell you the best way to get from one place on campus to another, just like Google Maps does for a city but built specifically for our college? It's a simple idea:

To solve this problem we thought of the campus as a graph. Every important location (library, main block, hostel, admin office etc.) is a node and the paths connecting these specific locations are the edges. With the whole campus represented in this way we can then use graph algorithms such as Dijkstra's algorithm to find the shortest or easiest path between any two points.

Honestly my favorite part of this project is that it is not just boring theory from a textbook. We actually get to use the stuff we learned in Data Structures and Algorithms class and turn it into something real and useful. It feels so cool knowing that this is not just for exams anymore it is something that could actually help someone who is lost on their first day just like I was.

State of the Art / Current solution

In a college, it can sometimes be challenging to find different locations, especially for freshers , visitors, and parents visiting. Most people depend on asking other students and teachers for directions. These ways are not much helpful and can be confusing if a person does not know the campus properly. Also, normal map apps do not always show the small roads, buildings, and paths inside a college campus.

To make this problem easier, our team decided to build a campus navigation system specifically for the college or any other institutions. In this system, all important places on campus and the paths connecting them can be marked. The user can select their current location and the place they want to go. The system will then show the shortest route between the two places. This will be beneficial for freshers, new faculty members and visitors to find places around the campus without having to ask someone for directions every time.

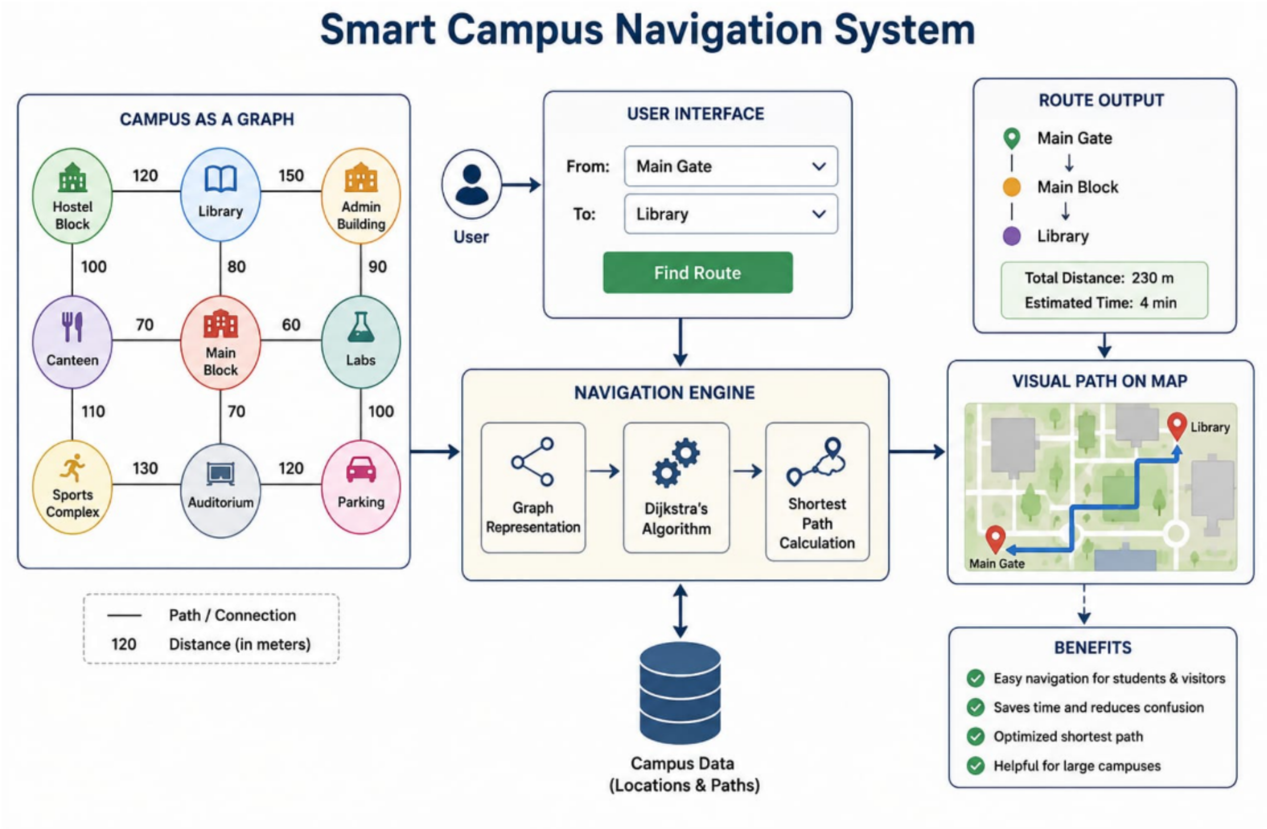
Project Goals and Milestones

Our project is about building a simple navigation system for our college campus that will make easier for students and visitors to find their way around. Basically, a user will select their starting location and choose where they want to go, and the system will calculate and provide them with the shortest route between the two locations . We've divided the project into weekly milestones so we can complete the project perfectly.
Week 1: We list out all the important places on campus such as library, hostels, canteen, departments and our day to day most important locations.
Week 2: We find out which of these places are actually connected to each other, and note down the paths and routes we can use in our daily lives .
Week 3: We converted this information into a graph structure such as nodes and paths as edges.
Week 4: Then our team Implement this graph in C++ and get a basic version of the navigation system running.
Week 5: We learn and use the Dijkstra's Algorithm so the system can measure the shortest path between any start and end point.
Week 6: Our team Test the project with different combinations of locations and points to see how it performs.
Week 7: We verify through the results, spot any errors or issues, and fix them.
Week 8: We finalise our project report, prepare the demo, and practice for the final presentation.

Project Approach

In this project we are going to design a Smart Campus Navigation System which will help people to find various places in our college campus. It will be mainly useful for students and visitors unfamiliar with campus. They won't have to keep bothering other people for directions. They can just pick where they are and where they want to go and the system will show a route.
We will model our college campus as a graph so that we can build the system. We will take important locations such as buildings and blocks as nodes. The edges will be the paths between these places. Then we will use shortest path algorithm to find a pathway between two locations given by user.
The project will be made using C++. We will include the campus graph, data structures and the shortest path algorithm in the program. We will also give the user an option to enter the starting point and destination. After making the program, we will test it by entering different locations and checking the routes given by the system. Some sample outputs and testing results will also be included in our project report. We will explain how our program works and how we have used the data structures and algorithm.

System Architecture



Project Outcome

The main result we want from this project is a Smart Campus Navigation System that can help students, teachers and visitors find their way around the college. It should make it easier for someone to go from one place to another, especially when they do not know the campus very well.
For making the system, we will show the college campus in the form of a graph. The important places in the campus will be taken as nodes and the paths between them will be the edges. The program will use a shortest path algorithm to find a suitable route from the place selected by the user to the destination.
Our final project will have the C++ program, the campus graph and the shortest path part. The user will be able to enter the starting place and the place they want to reach. The program will then show the route. We will test different places and routes to see if the answers are right. We will also add some sample outputs in our report. Along with this, we will explain the data structures we used, how the algorithm works, how we made the system and what we found during testing.c

Assumptions

We already know all the main spots on campus (like the library, canteen, classrooms, etc) and how they connect to each other.These places will be added to our system as different points, and the paths connecting them will show how we can travel from one place to another. We will also give a distance or cost to each path so that the program can compare the available routes.
We are assuming that the distances between the places will not change much during normal days. For now, our system will not deal with things like construction work, a blocked path, too much crowd or bad weather. These things can be considered later if we want to make the project more advanced. The user will also need to choose the starting point and destination from the places that are already added to the system. The main purpose of our project is to find a good and short route between these two places. For this, we will use a graph and a shortest-path algorithm. This will keep the project simple and allow us to focus mainly on how graph data structures can be used for campus navigation.

References

1. Jaibalaji, M. (2026). *3D Campus Navigation System: A Web-Based Interactive Graph with Shortest-Path Finding and Immersive Visualization.*
DOI: https://doi.org/10.55041/ijsmt.v2i5.089

2. Gbadamosi, O. A., Ilori, A. O., Oduwale, M. A., & Abiona, O. O. (2025). *Development of a Smart GIS-Based Campus Navigation and Shortest Path Routing System for OAUSTECH, Ondo State, Nigeria.*
DOI: https://doi.org/10.18488/79.v12i1.4656

3. Pilski, M., Mikułowski, D., & Terlikowski, G. (2023). *An Indoor Campus Navigation System for Users with Disabilities.*
Article: https://doi.org/10.34739/si.2023.29.06

4. Kunhoth, J., Karkar, A., Al-Maadeed, S., & Al-Ali, A. (2020). *Indoor Positioning and Wayfinding Systems: A Survey.*
Article: https://doi.org/10.1186/s13673-020-00222-0

5. Muhammad Riza Alifi, Ade Hodijah, Urip Teguh Setijohatmo, Sri Ratna Wulan, Hashri Hayati.
Article: https://e-jurnal.pnl.ac.id/JAISE/article/view/6839